Data Sources and Provenance

NJ TRANSIT River LINE Schedule Optimization (Team 22)

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Living document — last updated September 28, 2026 (public data pull added)

This document records where every number, file, and parameter used in the project comes from, how it was processed, and how much to trust it. Update it whenever a source is added or replaced. Links were accessed on September 28, 2026 unless noted.

Trust levels used below.

- **A** — official agency data or documents.
- **B** — reputable secondary sources, such as news reports quoting the agency, or Wikipedia citing the agency.
- **C** — third-party mirrors, or data that was machine-extracted.
- **P** — placeholder assumptions made by the team, to be replaced with client data.

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1 Schedule, stations, and running times

Item	Source and link	Trust	Project file
Stations, coordinates, track geometry, scheduled trips (2014)	NJ TRANSIT rail GTFS feed, service dates 2014-03-13 to 2014-09-08, route 16 "Riverline Light Rail." This is a third-party GitHub copy: `https://github.com/heychrisek/njtransit/tree/master/rail_data`. We used `stops.txt`, `trips.txt`, `stop_times.txt`, and shape 1404 in `shapes.txt` (33.59 mi).	C	`data/raw_gtfs_njt_rail_2014/`, `river_line_stops.csv`, `river_line_trips_2014.csv`, `river_line_stop_times_2014.csv`
Running time between each pair of stations	Derived from the 2014 GTFS stop times: the median scheduled minutes for each segment, by direction and day type. End-to-end time is 67 min.	C	`river_line_interstation_runtimes_2014.csv`
Vehicle blocks (train duties)	Derived from the GTFS `block_id` field. We saw 12 active blocks at the peak and 6 at the base.	C	`river_line_blocks_2014.csv`
Cross-check of the current timetable	The partial December 2024 weekday northbound timetable, read from `https://njtransh it.com/schedule/17/riverline-light-rail?date=2024-12-02&direction=1`. It shows 65–67 min end to end and the same 15/30-min pattern. This is an unofficial site.	C	`river_line_weekday_northbound_2024-12-02_partial.csv`
Current official schedule (to obtain)	NJ TRANSIT developer portal (free registration and license agreement): `https://www.njtransit.com/developer-tools`, and `https://developer.njtransit.com/registration/docs`. Also available through the Transitland feed page `https://www.transit.land/feeds/f-dr5-nj~transit~rail`. The timetable PDF is at `https://content.njtransit.com/sites/default/files/pdfs/rail/2025/07/10003/343.pdf` (linked from `https://www.njtransit.com/rline`).	A	not yet downloaded

Update (Sept. 28, 2026). The current feed has been retrieved (Section 8). Its scheduled end-to-end running time is still 67 minutes, the same as in 2014, so the model's running times remain valid. Slow orders are still not modeled.

2 Infrastructure and operating rules

Item	Source and link	Trust	Project file
Single and double track by milepost; siding names; facilities; 72 grade crossings; car capacity	NJDOT, *NJ Freight Guideway: RiverLINE fact sheet*, version 10/1/2021: `https://nj.gov/transportation/capital/guideway/pdf/F_RiverLINE.pdf`. It shows 12.7 mi of single track and 21.2 mi of double track, with sidings at Pennsauken, South Burlington, Florence, and Roebling. The facilities listed are the Camden Light Rail Complex, Burlington City Yard (MP 17), and the Trenton layover (MP 32.9).	A	`river_line_track_segments.csv`
Freight time separation	The same NJDOT sheet gives light rail 5:45 am–10:00 pm and freight 10:00 pm–5:45 am. The FRA waiver docket FRA-1999-6135 is at `https://www.govinfo.gov/content/pkg/FR-2007-01-12/pdf/E7-316.pdf`, and the horn-rule waiver FRA-2007-0030 at `https://www.govinfo.gov/link/fr/73/6260`.	A	model constant (day window)
Maximum speed; post-2024 speed restrictions	NTSB, operations factual report for investigation RRD25FR001 (Florence, NJ, Oct. 14, 2024): `https://data.ntsb.gov/Docket/Document/docBLOB?ID=19394735&FileExtension=pdf&FileName=Operations+Factual-RRD25FR001-Florence+NJ-Rel.pdf`. The investigation page is `https://www.ntsb.gov/investigations/Pages/RRD25FR001.aspx`. It gives a 65 mph maximum, and a 30 mph restriction extended to MP 23.9–27.4.	A	not yet in model

Processing. Railroad mileposts (0–33.9) were rescaled linearly onto GTFS shape distance (0–33.59 mi). This is an approximation. The team should obtain the actual track chart, with siding limits and switch locations, from NJ TRANSIT.

3 Fleet, operator, and reliability

Item	Source and link	Trust	Used as
Fleet: 20 Stadler GTW 2/6 cars, delivered 2002–2003	`https://en.wikipedia.org/wiki/River_Line_(NJ_Transit)`; `https://blog.pesky.moe/posts/2025-02-02-stadler-njt/`	B	fleet size 20
Car capacity: 90 seated, 184 total	NJDOT fact sheet (above)	A	comfort and crowding analysis

Item	Source and link	Trust	Used as
Operator change from Alstom to NJ TRANSIT, completed by Feb. 14, 2026	NJ TRANSIT press release, Sept. 3, 2025: https://www.njtransit.com/press-releases/nj-transit-and-alstom-joint-statement-river-line. NJT board minutes, Nov. 12, 2025: https://content.njtransit.com/sites/default/files/board/meeting_minutes/2025_11_12_OpenSess.pdf. Railway Gazette: https://www.railwaygazette.com/light-rail-and-tram/nj-transit-to-take-over-river-line-operations-from-alstom/69575.article	A	context
Maintenance contractor (Kinkisharyo); 14-car rehabilitation; fleet replacement RFP	https://www.njtransit.com/press-releases/nj-transit-announces-continued-river-line-customer-experience-and-communication; https://railway-news.com/nj-transit-outlines-ongoing-river-line-upgrades/ (Jan. 13, 2026); https://www.masstransitmag.com/rail/infrastructure/news/55342693/new-jersey-transit-nj-transit-nj-transit-advances-river-line-transition-with-system-and-equipment-upgrades	A/B	context; the long-term-out parameter
Miles between breakdowns: 22,337 (2022), 4,192 (2024)	ROI-NJ, Sept. 8, 2025: https://www.roi-nj.com/2025/09/08/transportation/nj-transit-will-take-over-river-line-light-rail-in-south-jersey/	B	**failure rate in the simulation** (4,192 mi)
Service cuts and heat-related failures, 2023–2024	Philadelphia Inquirer, Sept. 2023: https://www.inquirer.com/transportation/nj-transit-riverline-cancelations-delays-light-rail-20230920.html. News 12, 2024: https://newjersey.news12.com/nj-transit-to-provide-supplemental-bus-service-to-address-river-line-delays-and-cancellations. https://www.troysingleton.com/nj_transit_takes_control_of_troubled_river_line_light_rail_from_contractor	B	summer seasonal factor (qualitative)
NTD breakdowns, NJ TRANSIT light rail (NTD ID 20080)	https://data.transportation.gov/Public-Transit/2022-2024-NTD-Annual-Data-Breakdowns/amkt-4ehs. Correction (Sept. 28, 2026): "LR-PT" is Hudson–Bergen. The River LINE is reported separately as Hybrid Rail (mode YR, purchased transportation until Feb. 2026, directly operated after); see data_public/ntd_hybrid_rail/.	A	not used in the model
NTD agency profile, 2024	https://www.transit.dot.gov/sites/fta.dot.gov/files/transit_agency_profile_doc/2024/20080.pdf	A	context

Item	Source and link	Trust	Used as
NTD Revenue Vehicle Inventory, 2024 (to obtain)	https://www.transit.dot.gov/sites/fta.dot.gov/files/2025-10/2024%20Revenue%20Vehicle%20Inventory_250820.xlsx. Filter on the Stadler manufacturer to isolate the River LINE fleet.	A	not yet downloaded

4 Performance

Item	Source and link	Trust	Used as
On-time performance: 80.4% (Aug. 2023), 82% (July 2024), 84% (2025)	Inquirer (above); Troy Singleton (above); NJTOD newsletter: https://www.njtod.org/the-river-lines-untapped-potential/	B	**calibration target** for the simulation
On-time definition: within 6 minutes	NJ TRANSIT press release on its performance dashboard: https://www.njtransit.com/press-releases/nj-transit-launches-new-online-performance-dashboard	A	lateness threshold in the model
Monthly OTP dashboard (to extract)	https://www.njtransit.com/improve/on-time-performance/light-rail. The data loads by JavaScript and needs a browser.	A	not yet extracted
CEO monthly reports (to extract)	For example, https://content.njtransit.com/sites/default/files/board/edreport/edreport-September-2025.pdf	A	not yet extracted
Customer satisfaction surveys	https://www.njtransit.com/surveyresults	A	not yet extracted

5 Ridership and rider behavior

Item	Source and link	Trust	Used as
Weekday boardings by station, 2024 (total ≈5,573; Pennsauken TC missing)	https://en.wikipedia.org/wiki/River_Line_(NJ_Transit), which cites NJ TRANSIT	B	river_line_station_boardings_2024.csv; total demand (rounded to 5,600)
Ridership trend: 9,014 per weekday (2014), 6,330 (2019), about 5,350 (2022); 1.6M a year (2025)	Wikipedia (above); Inquirer (above); NJTOD (above)	B	context

Item	Source and link	Trust	Used as
2010 onboard survey: boardings and alightings by station, origins and destinations, trip purpose, access mode	DVRPC, *NJ Transit River LINE Survey*, publication 11057 (March 2013), survey of Nov. 17, 2010, 1,265 responses, first train to 2:30 pm: `https://www.dvrpc.org/reports/11057.pdf`	A	not yet used; basis for an origin–destination matrix
DVRPC transit ridership datasets	`https://catalog.dvrpc.org/dataset/transit-ridership-ntd`	A	not yet used
Hourly demand profile	**No source.** This is an assumed commuter-shaped profile, with 10% of daily boardings at 7–8 am and 9% each at 4–5 and 5–6 pm.	P	passenger wait and crowding

6 Weather

- NOAA GHCN-Daily station USW00014792, Trenton-Mercer Airport: `https://www.ncdc.noaa.gov/cdo-web/datasets/GHCND/stations/GHCND:USW00014792/detail`. Alternate station: USW00013739, Philadelphia International.
- Bulk download (daily TMAX, TMIN, PRCP, SNOW, AWND): `https://www.ncei.noaa.gov/access/services/data/v1?dataset=daily-summaries&stations=USW00014792,USW00013739&startDate=2015-01-01&endDate=2026-09-27&dataTypes=TMAX,TMIN,PRCP,SNOW,AWND&units=standard&format=csv`
- Trust A. Not yet downloaded. To be joined with the hold sheets to estimate how temperature affects failure rates.

7 Map layers (simulator only)

- Track line: GTFS shape 1404 (above).
- County and NJ/PA state lines: U.S. Census-derived `us-atlas` v3.0.1, 1:10M scale: `https://github.com/topojson/us-atlas`.
- Delaware River centerline: Natural Earth 10m rivers: `https://github.com/nvkelso/natural-earth-vector/blob/master/geojson/ne_10m_rivers_lake_centerlines.geojson`.
- Both map layers are coarse (trust C for placement). Use them for orientation only.

8 Public data retrieved September 28, 2026

The files below are in `data_public/`. Most hosts were not reachable from the cloud workspace. The data were fetched through a browser on the advisor's computer, reduced there to the River LINE subset, and checked with a 32-bit rolling hash on both ends. Every extracted file matched its source.

File	Source and link	Trust	Contents
gtfs_2026-09_weekday_timetable_raw.txt	NJ TRANSIT rail GTFS, release of Sept. 15, 2026, via the Mobility Database mirror (feed mdb-509): https://mobilitydatabase.org/feeds/gtfs/mdb-509. Producer URL: https://www.njtransit.com/rail_data.zip. River LINE is route 17 (RVLN).	A	87 weekday trips (Tue. Sept. 29, 2026); departure time at all 21 stations; block id
noaa_trenton_monthly_2015_2026.csv, noaa_trenton_daily_2026.csv	NOAA NCEI GHCN-Daily, station USW00014792, via https://www.ncei.noaa.gov/access/services/data/v1?dataset=daily-summaries	A	Monthly heat, freeze, rain and snow day counts; 2026 daily values
ntd_njt_lightrail_monthly_2015_2026.csv	FTA NTD Complete Monthly Ridership, https://data.transportation.gov/d/8bui-9xvu, NTD ID 20080, mode LR	A	Monthly trips, vehicle revenue miles and hours, vehicles in maximum service (DO and PT)
ntd_njt_lightrail_breakdowns_2022_2024.csv, ..._metrics_..., ..._service_by_period_..., ntd_njt_all_modes_metrics_2024.csv	NTD annual datasets: breakdowns https://data.transportation.gov/d/amkt-4ehs; metrics https://data.transportation.gov/d/ekg5-frzt; service by time period https://data.transportation.gov/d/wwdp-t4re	A	Failures, cost per revenue hour, trains in operation by period; River LINE is reported as Hybrid Rail (YR), separate from HBLR
osm_river_line_route_ways.csv, osm_derived_single_track_intervals.csv	OpenStreetMap route relations 206683 (to Trenton) and 10213648 (to Camden) and their member ways, via the Overpass API (https://overpass-api.de)	C	Posted speed per way; ways used by both directions (single track)
dvrpc_2010_survey_summary.csv	DVRPC publication 11057 (2013): https://www.dvrpc.org/reports/11057.pdf	A	Trip purpose, access/egress mode, fare type
gtfs_2014_river_line_shape_camden_to_trenton.csv	2014 GTFS shape 1404, reversed to run Camden to Trenton	C	Track polyline with mileage

9 Model parameters and where each comes from

Parameter	Value	Trust	Basis
Fleet size	20 cars	B	Wikipedia; blog
Cars in long-term overhaul	3	P	Guess, informed by the 14-car rehabilitation program
Miles between breakdowns	4,192	B	ROI-NJ (2024 figure)
Seasonal breakdown multipliers	W 0.85, Sp 1.25, Su 0.65, F 1.0	P	Client notes (heat worst, spring best) and the heat-failure literature in the review
Background late probability	3–9% by season	P	Tuned so on-time $\approx$ 84–86% under the fixed schedule
Mean repair time	12 days	P	Tuned so mean availability is about 9 cars, consistent with "fewer than 8 often"
Rescue and blockage	10 min dispatch, 25/15 mph, 35 min block	P	Guesses; ask the client
Minimum layover	5 min	C	2014 GTFS blocks show 5–9 min
Meet clearance buffer	1 min	P	Chosen so the model reproduces the 8-train/20-min and 6-train/30-min schedules
Weekday boardings	5,600	B	2024 station boardings
Car capacity	90 seated / 184 total	A	NJDOT
Operating cost per train-hour	\$400 (range \$150–800)	P	Assumption, bracketed by NTD 2024 light rail cost per revenue hour: \$560 (DO), \$809 (PT)
Value of rider time	\$13.60/h; waiting $\times$2	A	USDOT 2016 VTTS guidance, local personal travel, 2015 dollars
Bus-bridge cost	\$170/bus-hour	P	Between NTD 2024 NJT bus \$126 (PT) and \$214 (DO)

10 Data to request from NJ TRANSIT

1. Hold sheets since Feb. 14, 2026: cars available and down, recorded at 6 am and 6 pm. This is the key input.
2. Work orders and failure logs by car: date, failure type, mileage, and days out of service.
3. Delay, cancellation, and incident logs by train, with causes.
4. The current slow-order list and the track chart: siding limits, switches, and speed limits.
5. Ridership: automatic passenger counts (if equipped), ticket machine and mobile-app sales or activations by station and hour, and any recent onboard survey. Ask DVRPC for the raw

2010 survey.

6. Minimum layover times, consist rules, crew constraints, and where protect cars can be staged.
7. The names of NJT's scheduling system, maintenance system, and dashboard platform, and access to the River LINE real-time feed.